RallyScope

INTERACTIVE BADMINTON PERFORMANCE & RIVALRY ANALYTICS

WEEK 5 • MINI-PROJECT 2 **ADVAIT JISHNANI** NET ID: AJ4700

01 Project overview

Goal. Build an interactive badminton dashboard that connects comeback performance, singles-versus-doubles competitiveness, and close head-to-head rivalries. Users will explore all three questions through coordinated views.

Why badminton. As a **professional badminton player**, this topic really spoke to me. I have experienced how quickly momentum can shift and how intense rivalries can become. I wanted to explore those patterns through data while combining my interest in sports and analytics.

Technology choice. After working with Tableau, I wanted to learn a new tool. Plotly Dash is a good opportunity to build interactive views using Python and learn it from scratch.

Selected layout. Balanced Overview: two comparison charts across the top, one wide rivalry chart below, and shared filters above all three.

02 Public dataset

FIELD	DETAIL
Dataset	BWF Match Data (public CSV files, one row per completed match)
Source	github.com/SahilMotyar/bwf-match-data
Coverage	105,147 matches, 608 elite tournaments, Jan 2007–Jul 2026
Disciplines	Men's/Women's Singles, Men's/Women's Doubles, Mixed Doubles
Main fields	date, discipline, tournament, tier, round, team1, team2, winner, score
Derived fields	opening-game loser, comeback outcome, deciding-game flag, game-point margin, head-to-head meetings and win balance

03 Research questions & measures

RQ1 Greatest comeback players

Which players or pairs most consistently win after losing the opening game, and how does this vary by discipline and tournament level?

Metric: Comeback success rate = matches won after dropping Game 1 / matches in which Game 1 was lost. Require at least 10 opportunities per player or pair.

RQ2 Singles vs doubles competitiveness

Do singles and doubles matches differ in how often they reach a deciding third game and in their game-score margins?

Metric: Deciding-game rate = three-game matches / completed best-of-three matches, compare MS, WS, MD, WD, XD. Average per-game winning margin provides a second measure.

RQ3 Closest rivalries

Which players or pairs have the most evenly divided head-to-head records, among opponents who have met at least six times?

Metric: Rivalry parity = $100\% - 2 \times |\text{Side A win share} - 50\%|$. A perfect split scores 100%, bubble size shows number of deciding games in the rivalry.

04 Coordinated dashboard interaction

View A · Comeback leaders. Ranked horizontal bars, clicking a competitor filters matches involving that player or pair.

View B · Competitiveness. Deciding-game rates for all five disciplines, with average game-score margin in tooltips, clicking a discipline filters the other views.

View C · Rivalries. Bubble chart of meeting count vs. parity, selecting a rivalry filters the other charts to those two sides.

Shared behavior. Year, discipline, and tournament level filters apply across the dashboard. Every chart also supports hover details and clearing selections.

Data cleaning and scope

I will parse the ordered game scores, exclude incomplete or ambiguous score strings from rate calculations, and standardize player/pair identifiers. Only completed, score-valid best-of-three matches will enter deciding-game and comeback metrics. Tournament rounds will be mapped to an explicit order (R32 → R16 → QF → SF → Final).

All wireframe values on the next pages are illustrative placeholders, they are not findings from the dataset.

05 Dashboard sketches · selected layout

Selected: Balanced Overview. The top row makes comeback leaders and competitiveness easy to compare, the bottom panel gives rivalries more horizontal space for exploration.

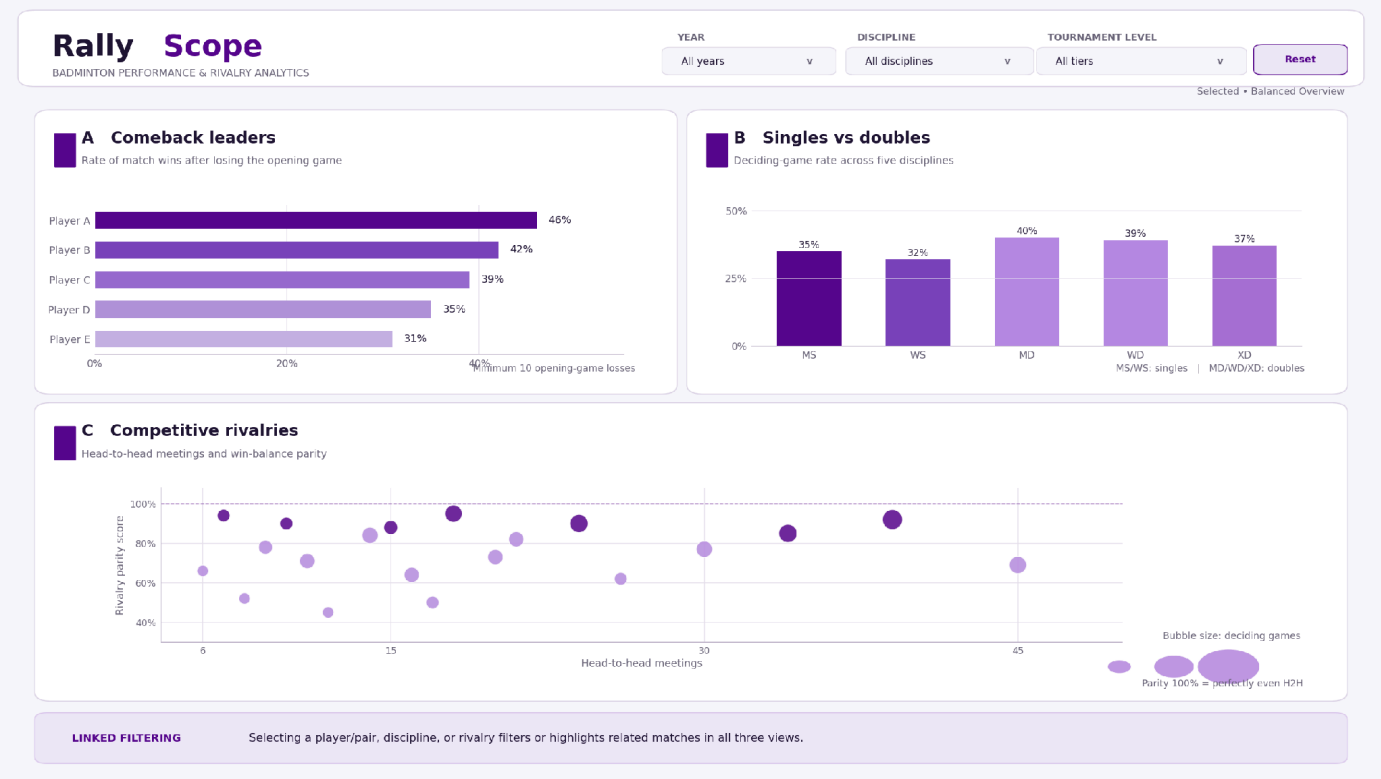


Figure 1. Balanced Overview (selected). All names and percentages shown are illustrative, not observed results.

Why this layout?

Gestalt proximity and enclosure. Each research question has a distinct card. Related charts share a grid, while the filter strip is visually separated above them.

Visual hierarchy. The rivalry panel receives the full lower row because it needs space for two quantitative axes, bubble sizes, and detailed tooltips.

Cross-filtering. Selections in any chart will highlight or filter related matches in the other two charts, a single Reset control clears selections.

06 Alternative layouts considered

I compared two other arrangements before choosing the Balanced Overview. Both retain the same three linked research views, but shift which question receives the most space.

Alternative A · Rivalry-focused

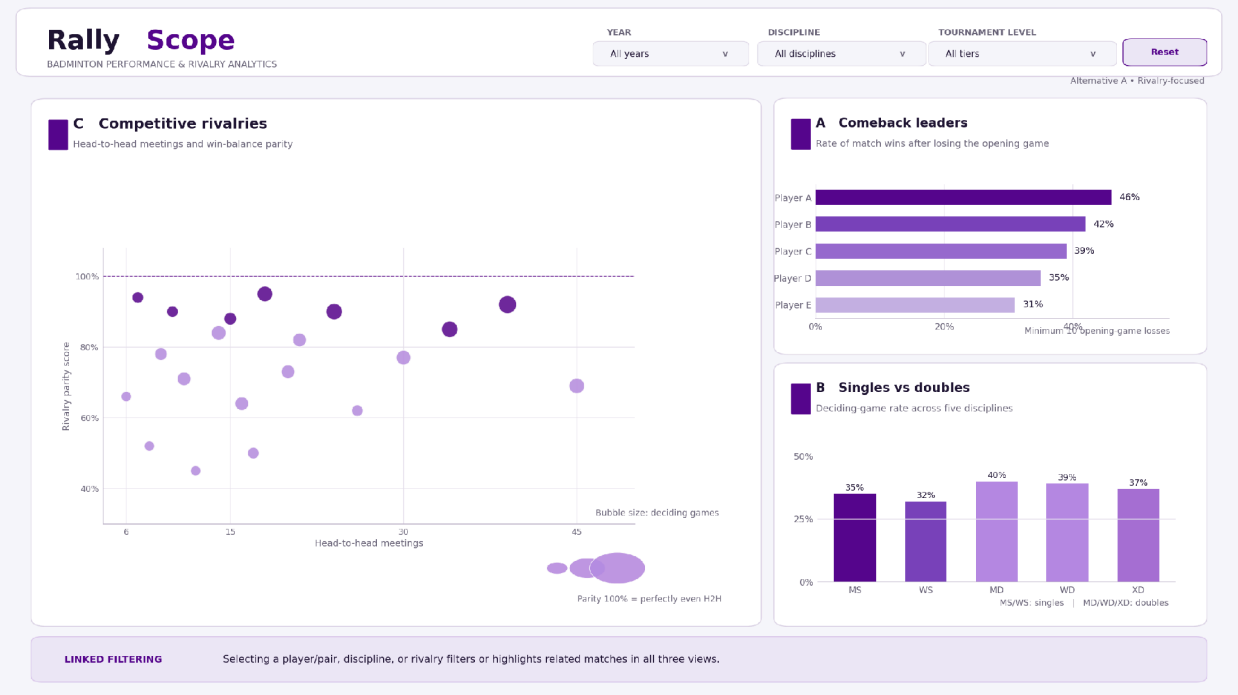


Figure 2. Makes the rivalry plot dominant, with two smaller supporting comparisons.

Alternative B · Trend-focused

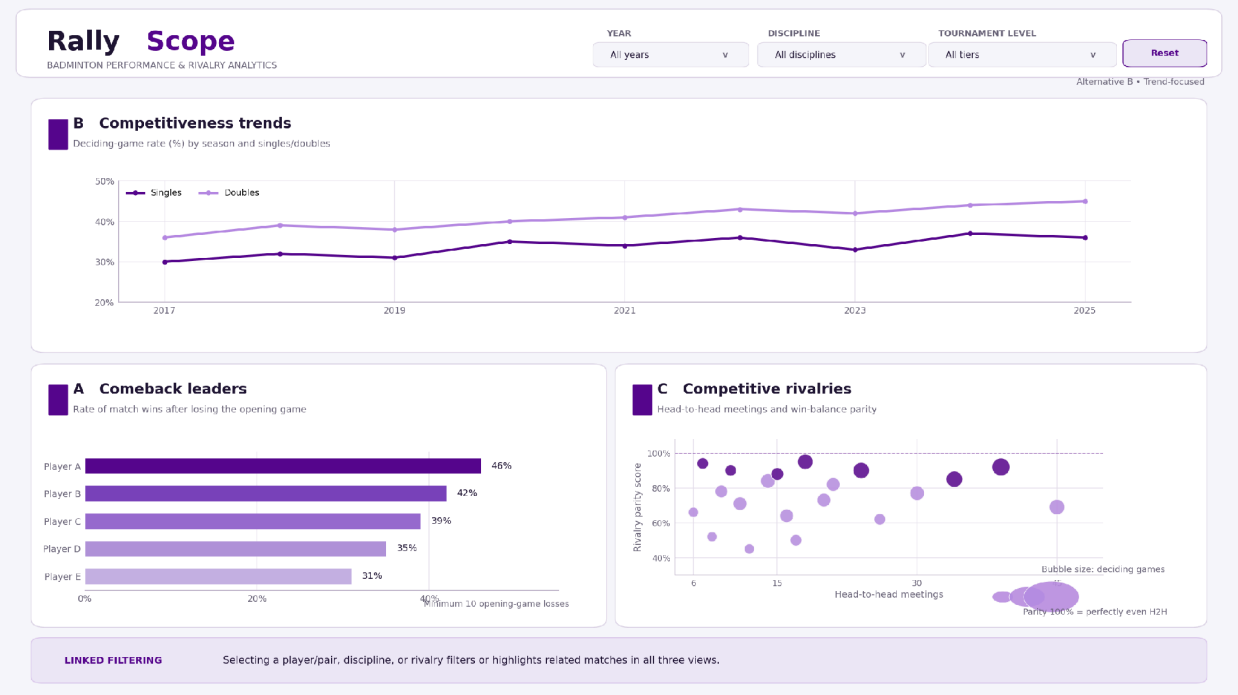


Figure 3. Emphasizes season trends, with comeback and rivalry charts beneath.

Main reason for not selecting these:
-> A is less balanced across the three questions, B emphasizes year-by-year trends more than the main singles-versus-doubles comparison. The chosen layout communicates all three questions more evenly.

07 Channel justification

The table describes the marks and channels in the selected Balanced Overview wireframe. Quantitative variables use aligned position and length, color and labeling identify categories, bubble area carries supplementary match-volume context.

Variable	Type	Mark	Channel	Principle	Justification
Player / pair	Nominal	Bar	Y label	Expressiveness	Identifies a competitor in the ranked list.
Comeback success rate	Quantitative	Bar	Length (X)	Effectiveness	Aligned bar lengths compare percentages.
Discipline	Nominal	Bar	X category + hue	Expressiveness	MS, WS, MD, WD, XD are labeled, hue helps group singles and doubles.
Deciding-game rate	Quantitative	Bar	Height (Y)	Effectiveness	Position represents the fraction requiring three games.
H2H meetings	Quantitative	Point	X position	Effectiveness	Position exposes rivalry sample size.
Rivalry parity	Quantitative	Point	Y position	Effectiveness	Higher values mean more even head-to-head splits.
H2H deciding games	Quantitative	Bubble	Area / size	Effectiveness	Area adds the number of deciding-game meetings.
Tournament tier	Ordinal	Filter	Ordered menu	Convention	Tour levels are ordered within each competition era.
Year	Temporal	Filter	Year range	Convention	Chronological range applies to all charts.

References and responsible use of AI

Dataset: Sahil Motyar, BWF Match Data (2007–2026), GitHub. https://github.com/SahilMotyar/bwf-match-data (accessed October 2026).

AI assistance: I used ChatGPT to brainstorm and refine research questions. Gemini assisted with formatting this document. I reviewed and selected the project direction, layout, creating the wireframes and deciding the encoding choices myself.